

SECURITY OPERATIONS CENTER (SOC) ANALYSIS REPORT

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Target: SOC Sample Logs Analysis — File: SOC_Task2_Sample_Logs

Tools Used: Splunk (SPL Queries, Dashboards)

Track Code: FUTURE_CS_02

1. Background

As part of Task 2, I was assigned to review and analyze the dataset SOC_Task2_Sample_Logs. The goal was to simulate a real SOC workflow:

- Ingest logs into Splunk
- Identify suspicious events
- Prioritize incidents
- Suggest immediate and long-term responses

2. Investigation Process

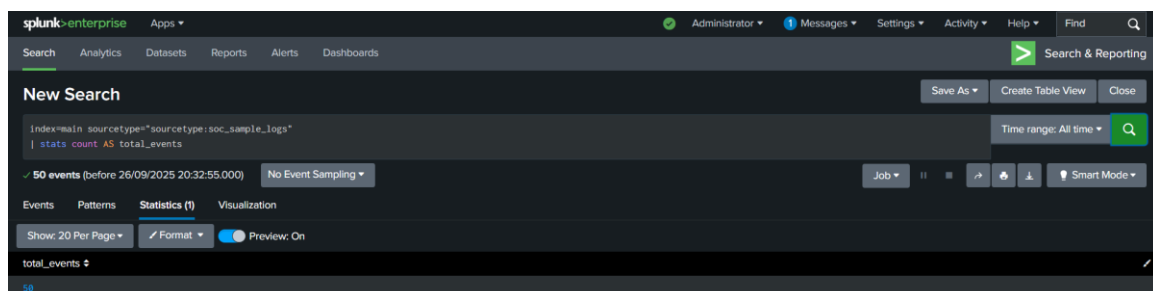
1. Uploaded the dataset into Splunk (index=main).
2. Queried logs for counts, user activity, and malware detection.
3. Correlated login attempts with outcomes (failed → success).
4. Highlighted abnormal IP activity.
5. Mapped events into a timeline to understand attack progression.

3. Observations

- Total Logs Analyzed: 50

Event Breakdown:

- 12 → Connection Attempts
- 11 → Login Success
- 11 → File Access
- 11 → Malware Detected
- 5 → Login Failures



New Search

index=main sourcetype="sourcetype:soc_sample_logs"

| stats count BY action

| sort - count

Time range: All time

50 events (before 26/09/2025 20:33:20.000) No Event Sampling

Events Patterns Statistics (4) Visualization

Show: 20 Per Page Format Preview: On

action	count
login	16
connection	12
file	11
malware detected	11

Malware Findings:

- 6 Trojans
- 2 Rootkits
- 1 Ransomware behavior
- 1 Worm infection attempt
- 1 Spyware alert
- *(Impacted users: bob, alice, eve equally affected)*

New Search

index=main sourcetype="sourcetype:soc_sample_logs" action="malware detected"

| stats count BY threat

| sort - count

Time range: All time

11 events (before 26/09/2025 20:36:01.000) No Event Sampling

Events Patterns Statistics (5) Visualization

Show: 20 Per Page Format Preview: On

threat	count
Trojan	6
Rootkit	2
Ransomware	1
Spyware	1
Worm	1

C. Suspicious IP Activity

IP 203.0.113.77 recorded 15 events — significantly higher than other IPs, suggesting probing or malicious behavior.

index=main sourcetype="sourcetype:soc_sample_logs" action="malware detected" threat=Worm

Time range: All time

1 event (before 26/09/2025 20:37:06.000) No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect

1 millisecond per column

Format Show: 20 Per Page View: List

Time	Event
03/07/2025 05:06:14.000	2025-07-03 05:06:14 user=bob ip=203.0.113.77 action=malware detected threat=Worm Infection Attempt host = SOHAILSHAIK source = SOC_Task2_Sample_Logs (2).txt sourcetype = sourcetype:soc_sample_logs

D. Failed → Success Login Sequences

bob experienced a login failed at 2025-07-03 04:47:14 from 10.0.0.5, followed by a login success at 2025-07-03 05:04:14 from 192.168.1.101 (delta ≈ 17 min). This is suspicious and may indicate credential compromise.



New Search [Save As] [Create Table View] [Close]

Time range: All time [Search]

```
index=main sourcetype=sourcetype:soc_sample_logs
| rex field=_raw "(?<td>\d{4})~\d{2})~\d{2})\s*\|\\suser=(?<user>[^\s|]+)\s*\|\\sip=(?<ip>[^\s|]+)\s*\|\\saction=(?<action>[^\s|]+)?(?<threat>[^\s|]+)?)"
| eval _time = strptime(ts, "XY-mm-dd HH:MM:SS")
| sort @ user _time
| streamstats current=f last(action) as prev_action last(ip) as prev_ip by user
| where prev_action="login failed" AND action="login success"
| table _time user ip action prev_ip prev_action
```

3 events (before 26/09/2025 20:38:15.000) [No Event Sampling]

Events Patterns Statistics (3) Visualization

Show: 20 Per Page [Format] [Preview: On]

_time	user	ip	action	prev_ip	prev_action
2025-07-03 04:47:14	alice	198.51.100.42	login success	203.0.113.77	login failed
2025-07-03 05:04:14	bob	192.168.1.101	login success	10.0.0.5	login failed
2025-07-03 05:18:14	charlie	172.16.0.3	login success	198.51.100.42	login failed

E) Suspicious Alerts & Prioritization

- **Ransomware Behavior** — user=bob at 2025-07-03 09:10:14
Priority: High
Suggested action: Isolate bob's host, collect disk image, run EDR, block suspect IPs.
- **Rootkit Signature** detected for alice and eve
Priority: High
Suggested action: Quarantine hosts, collect memory images, escalate to IR team.
- **Multiple Trojan detections (bob, alice, eve)**
Priority: Medium-High
Suggested action: Identify common file/hash, scan endpoints, check for lateral movement.
- **Worm Infection Attempt for bob** from 203.0.113.77
Priority: Medium
Suggested action: Monitor for scanning behavior, block IP if confirmed malicious.
- **Failed→Success** login sequence for bob
Priority: Medium-High
Suggested action: Force password reset, review sessions, investigate account activity.

Alert	Priority
Ransomware Behavior	High
Rootkit Signature	High
Trojan Detected	Medium-High
Worm Infection Attempt	Medium
Failed→Success Login (bob)	Medium-High

4. Incident Prioritization

Incident Type	Severity	Action Required
Ransomware activity (bob)	Critical	Host isolation, disk image collection
Rootkits (alice, eve)	Critical	Quarantine, memory capture, IR escalation
Trojan infections	High	File analysis, endpoint scans
Worm attempt (external IP)	Medium	IP block, scan for spreading activity
Suspicious login (bob)	High	Password reset, review access logs

5. Risk Assessment

- High Risk: Malware (ransomware/rootkits) spreading internally
- High Risk: Credential compromise (bob's login anomaly)
- Moderate Risk: Worm attempts and external IP scanning
- Ongoing Exposure: Lack of strong authentication controls

6. Recommended Response

Immediate Steps:

- Isolate infected hosts (bob, alice, eve)
- Reset and secure compromised accounts
- Block malicious IPs

Short-Term Steps:

- Enforce MFA for all user accounts
- Conduct targeted endpoint scans
- Enhance Splunk alerting thresholds

Long-Term Steps:

- Deploy EDR for continuous monitoring
- Conduct user awareness training
- Establish stronger incident response playbooks

7. Conclusion

The SOC Task 2 dataset revealed active ransomware, rootkits, and trojans, with clear signs of credential compromise. The unusual activity from IP 203.0.113.77 further strengthens the likelihood of external intrusion attempts.

Result: The environment is under active threat and requires urgent containment, followed by strategic improvements in monitoring and authentication security.



8. Evidence — SPL Commands

```
index=main sourcetype="soc_sample_logs" | stats count
```

```
index=main sourcetype="soc_sample_logs" | stats count BY action
```

```
index=main sourcetype="soc_sample_logs" action="malware detected" | stats count BY threat
```

```
index=main sourcetype="soc_sample_logs" action="malware detected" | stats count BY user
```

```
index=main sourcetype="soc_sample_logs" | stats count BY ip
```